

Formula 1 Dashboard Analysis

Business Intelligence project

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Introduction

Formula 1 is one of the world's most data-driven sports, generating extensive information on races, drivers, constructors, pit stops, and championship standings. However, raw data alone does not provide meaningful insights.

To address this challenge, we developed an interactive Power BI dashboard that transforms Formula 1 race data into clear visualizations, enabling users to identify performance patterns, compare competitors, and make data-driven decisions.

Project Objectives

The project was designed to:

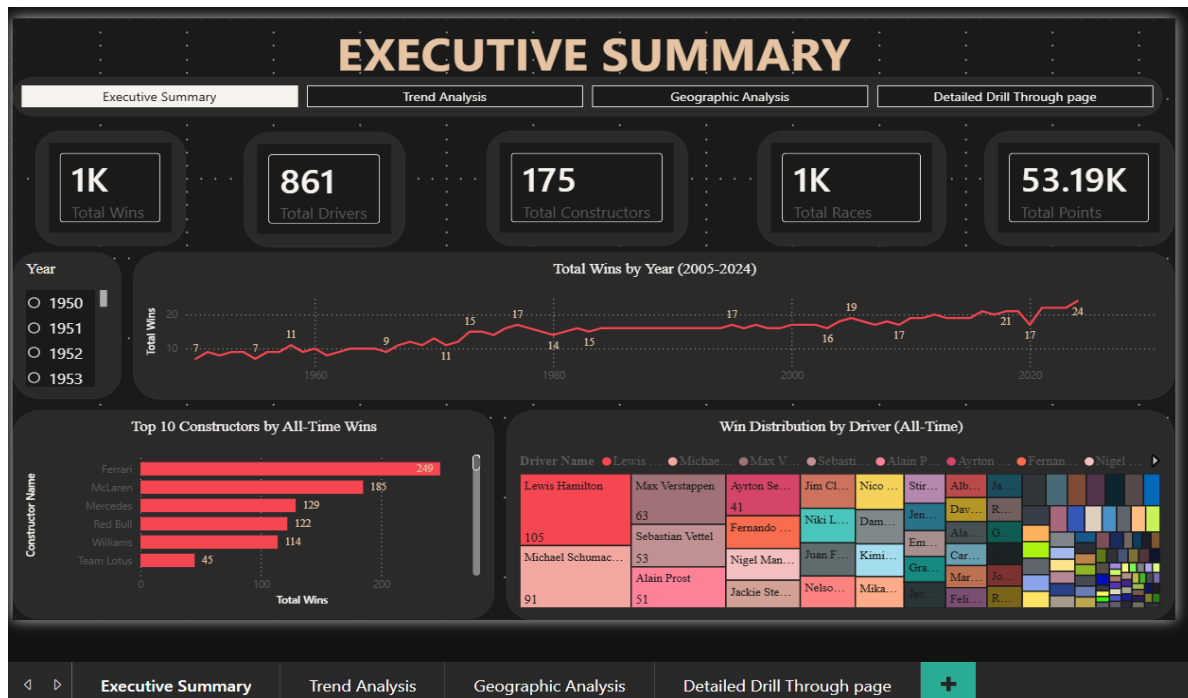
- Analyse Formula 1 performance using interactive dashboards.
- Identify drivers with the highest cumulative pit stops.
- Explore historical trends across seasons.
- Compare performance geographically.
- Enable detailed investigation through drill-through analysis.

Dataset Overview

 circuits	Microsoft Excel Comma Separ...
 constructor_results	Microsoft Excel Comma Separ...
 constructor_standings	Microsoft Excel Comma Separ...
 constructors	Microsoft Excel Comma Separ...
 driver_standings	Microsoft Excel Comma Separ...
 drivers	Microsoft Excel Comma Separ...
 lap_times	Microsoft Excel Comma Separ...
 pit_stops	Microsoft Excel Comma Separ...
 qualifying	Microsoft Excel Comma Separ...
 races	Microsoft Excel Comma Separ...
 results	Microsoft Excel Comma Separ...
 seasons	Microsoft Excel Comma Separ...
 sprint_results	Microsoft Excel Comma Separ...
 status	Microsoft Excel Comma Separ...

Dataset Characteristics

- Formula 1 historical race data
- Driver information
- Constructor statistics
- Race circuits
- Countries
- Championship points
- Pit stop records



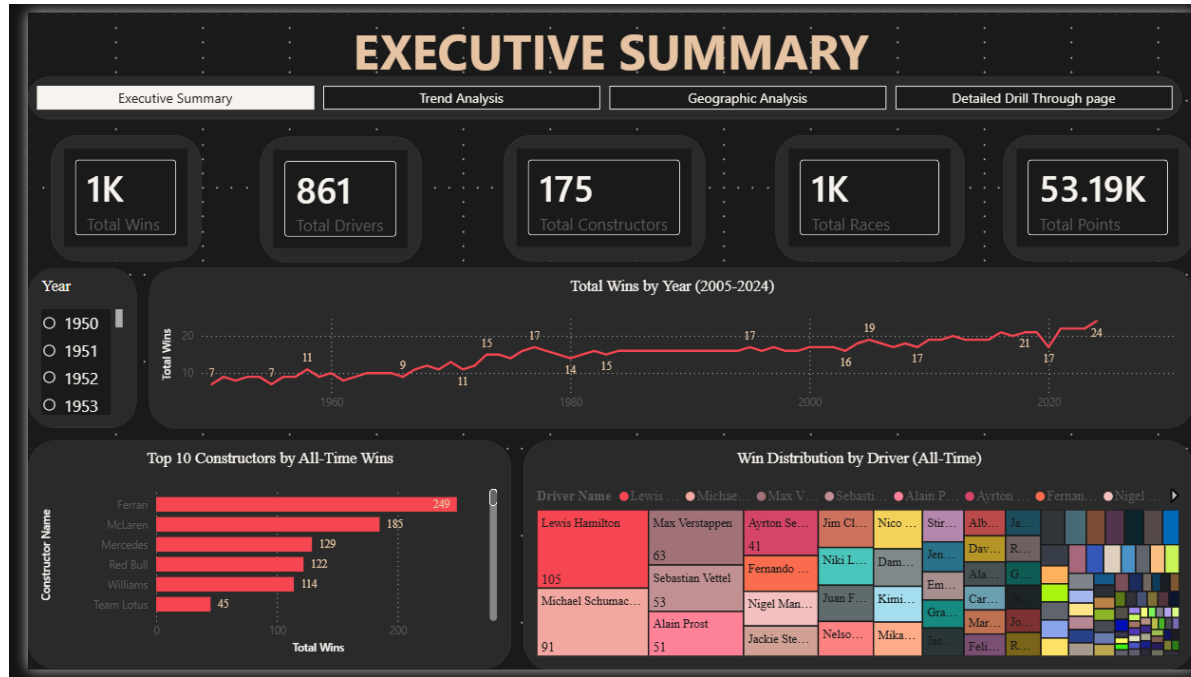
Dashboard Overview

The dashboard consists of four analytical pages:

- Executive Summary
- Trend Analysis
- Geographic Analysis
- Drill-through Analysis

Each page answers a different analytical question while remaining fully interactive through filters and slicers.

Executive Summary



Purpose

Provides a high-level overview of Formula 1 performance.

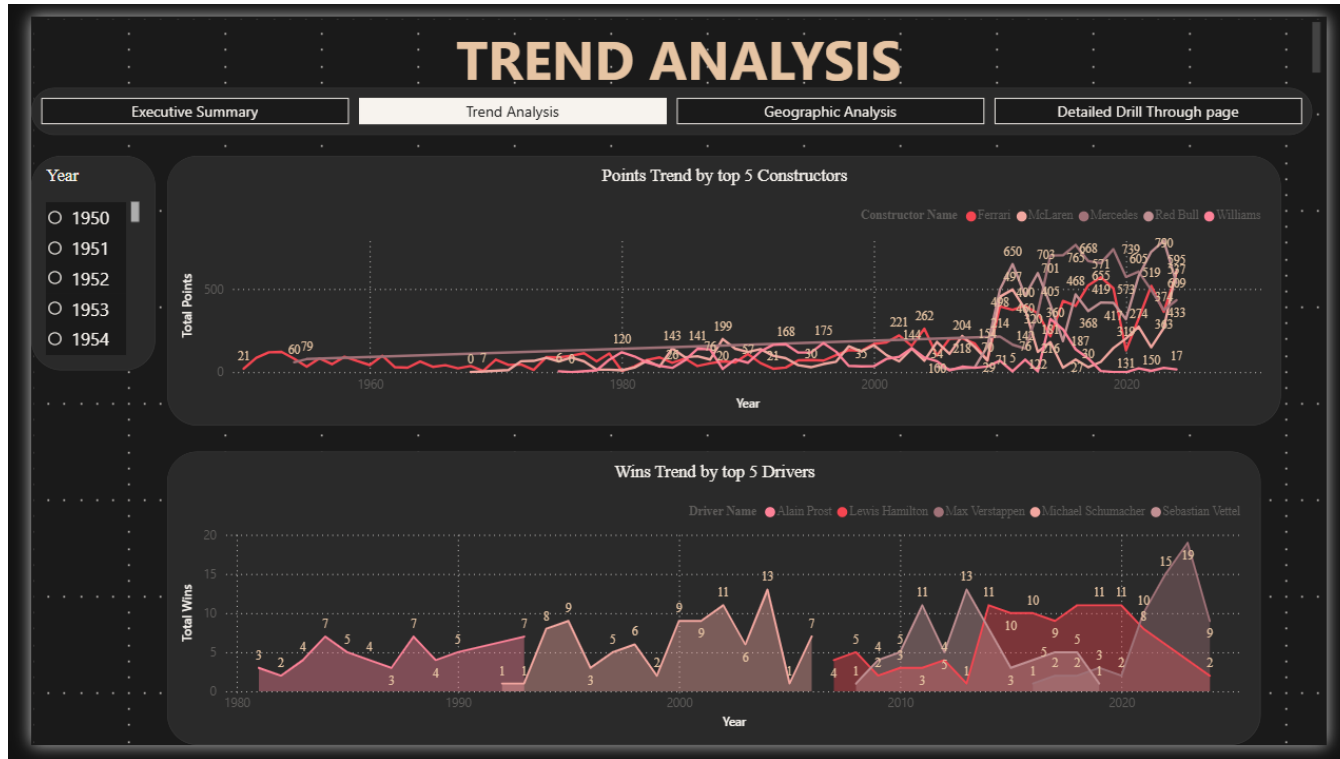
Business Value

Allows users to quickly understand overall performance before exploring detailed analysis.

Key Visualizations

- KPI Cards
- Driver Rankings
- Bar Charts
- Tree map
- Line Charts
- Interactive Slicers

Trend Analysis



Purpose

Examines how Formula 1 performance changes over time.

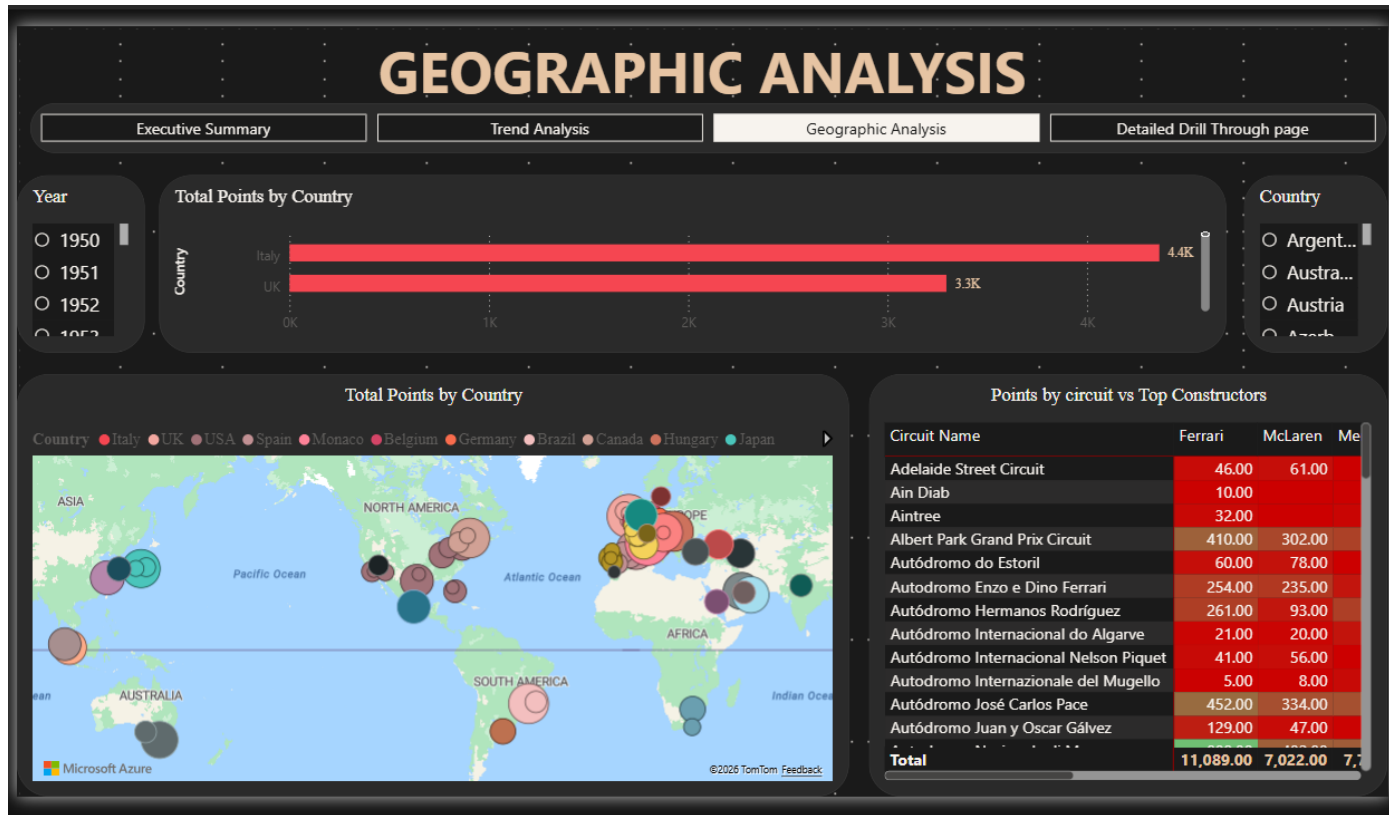
Visualizations

- Line Chart
- Area Chart
- Seasonal Analysis

Insights

- Identifies long-term performance trends.
- Highlights seasonal changes.
- Supports forecasting and strategic analysis.

Geographic Analysis



Purpose

Visualizes Formula 1 performance across countries and regions.

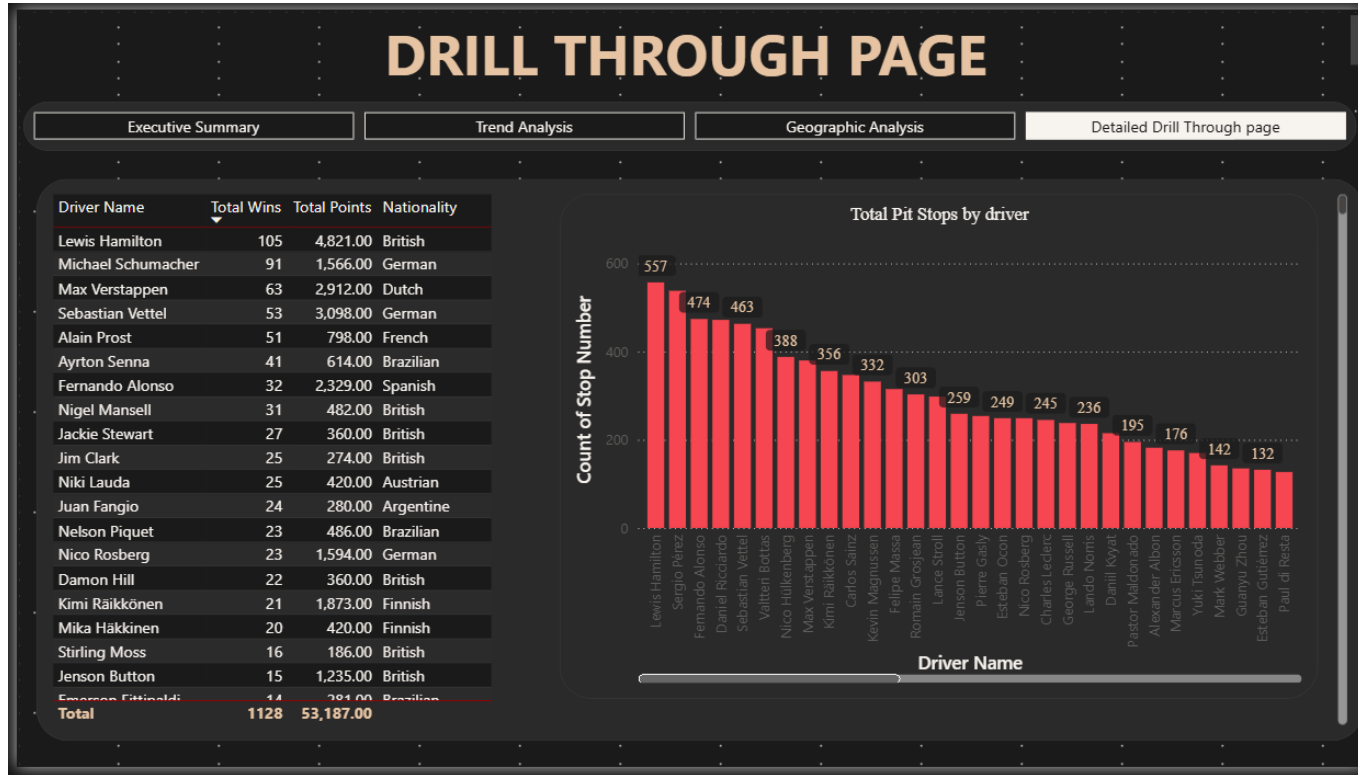
Visualizations

- Azure Map
- Bar Chart
- Matrix Table

Insights

- Identifies countries contributing the highest points.
- Compares regional performance.
- Highlights geographical patterns.

Drill-through Analysis



Purpose

Provides detailed transaction-level analysis.

Visualizations

- Detailed Tables
- Combo Charts
- Interactive Filters

Benefits

Users can investigate individual drivers, races, and performance records without overcrowding the main dashboard.

Key Findings

After analysing the dashboard, several important insights emerged:

- Drivers differ significantly in cumulative pit stop totals.
- Historical trends reveal changes in race strategies across seasons.
- Geographic analysis highlights countries contributing the largest share of Formula 1 success.
- Interactive dashboards improve exploration compared to static reports

Recommendations

Based on the dashboard analysis, we recommend:

- Regular monitoring of driver performance using cumulative statistics.
- Expanding the dashboard with predictive analytics.
- Incorporating additional race metrics for deeper analysis.
- Using interactive dashboards to support strategic decision-making.

Conclusion

The Power BI dashboard successfully converts complex Formula 1 data into meaningful business insights.

By combining KPI monitoring, trend analysis, geographic reporting, and drill-through capabilities, the dashboard provides an effective platform for analysing driver performance and supporting informed decision-making.

The project demonstrates how Business Intelligence tools transform raw data into valuable information for both analysts and decision-makers.